Lone Star Rides Rentals Management System

Phase 1: Planning, Design, and Proposal Submission

Course: Web Application Development

Institution: Houston Christian University

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Submitted by Group 10:

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1) Project Proposal

Project Title: Lone Star Rides Rentals Management System (LSR)

1. Purpose

The **Lone Star Rides Rentals Management System** is designed to digitize and streamline the operations of a car rental business. The primary goal is to replace manual booking processes with a dynamic web interface that handles real time vehicle availability, customer reservations, and automated billing. This system aims to reduce administrative overhead while providing customers with a frictionless, 24/7 booking experience.

2. Target Users

- **Casual Renters:** Individuals looking for short-term vehicle rentals for travel or personal use.
- **Corporate Clients:** Business users requiring long-term or fleet-based rental agreements.
- **Fleet Managers/Administrators:** Staff members responsible for inventory updates, maintenance scheduling, and financial reporting

3. Main Features

- **Dynamic Fleet Catalog:** A searchable gallery of available vehicles filtered by make, model, price, and fuel type.
- **Real-Time Booking Engine:** A calendar-based reservation system that prevents double-booking and calculates costs based on rental duration.
- **User Dashboard:** A personal space for customers to view upcoming reservations, download invoices, and manage their profiles.
- **Responsive Search & Filters:** Users can find vehicles based on specific criteria (e.g., SUVs, Luxury, Economy).
- **Automated Email Notifications:** System-generated confirmations for bookings, cancellations, and return reminders.

4. User Roles and Permissions

This Table will explain the roles and the permissions for each roles and what features it can have.

Roles	Permissions
Guest	Can browse the vehicle catalog, view pricing, and create a new account.
Registered Customer	Can book vehicles, manage their own profile, view booking history, and cancel reservations.
Administrator	Can add/edit/remove vehicles from the fleet, manage all user accounts, view revenue reports, and override booking statuses.

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Course: Web Application Development

Project Title: Lone Star Rides Rentals Management System (ARMS)

Topic: Cloud-based Vehicle Rental and Fleet Management

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5. External API Integration

To enhance the functionality and realism of the application, the system will integrate the **Google Maps Platform API**.

- **Implementation:** The API will be used to provide a "Branch Locator" feature, allowing users to see pickup and drop-off locations on an interactive map. It will also utilize the **Distance Matrix API** to calculate mileage-based fees if a user chooses to drop off a vehicle at a different branch than the pickup location.

2) System Planning

1. List of Pages

- **Public Pages (Guest access)**

Pages	Purposes
Home / Landing Page	Featured vehicles, CTA to browse or sign up
Vehicle Catalog	Browse/search all available cars with filters
Vehicle Detail Page	Full info on a single car (photos, specs, pricing, availability)
Branch Locator	Google Maps integration showing pickup/drop-off locations
Login Page	Sign in for existing users
Register Page	New account creation

- **Customer Pages (Registered users)**

Pages	Purposes
Customer Dashboard	Overview of upcoming & past reservations
Book a Vehicle	Calendar-based reservation form with cost calculator
Booking Confirmation	Summary + email trigger after successful booking
My Reservations	View, manage, or cancel active bookings
My Profile	Edit personal info, license details, password
Invoice / Receipt	Downloadable billing summary per booking

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- **Admin Pages (Administrator Only)**

Pages	Purpose
Admin Dashboard	Revenue snapshot, active bookings, fleet status
Fleet Management	Add, edit, or remove vehicles
Booking Management	View all bookings, override statuses
User Management	View/edit/deactivate user accounts
Reports	Revenue reports, booking history exports

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2. Navigation Structure

The system exposes three distinct navigation contexts depending on the user's role and login state. Each context shows only the pages relevant to that role, keeping the interface clean and secure.

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- **LSR Website**
- **|**
- **|---- [Public Nav]**
- **| |- Home**
- **| |- Browse Vehicles**
- **| |-Branch Locator**
- **| |-Register**
- **|**
- **|---- [Customer Nav – Logged in]**
- **| |- Home**
- **| |- Browse Vehicles**
- **| |- My Dashboard**
- **| |- My Profile**
- **| |- Logout**
- **|**
- **|__ [Admin Nav – admin login]**
- **|- Admin Dashboard**
- **| - Fleet Management**
- **|- Booking Management**
- **|- User Management**

- |- Reports
- |- Logout
-

3. Main Modules / Sections

Module 1 – Authentication Module

- Handles registration, login, logout, and session management. Differentiates access between Guest, Customer, and Admin.

- **Module 2 – Fleet Catalog Module**
- Displays all vehicles with search and filter functionality (by type, price, fuel, availability). Powered by the database and visible to all users.

- **Module 3 – Booking/Reservation Module**
- The core engine of the system. Allow customers to select a vehicle, choose pickup/drop-off dates and locations, and calculate the total cost. Prevent double-booking by checking real-time availability.

- **Module 4 – User Dashboard Module**
- A personalized portal for customers to track upcoming reservations, download invoices, and edit their profiles.

- **Module 5 – Admin Control Module**
- Restricted to administrators. Covers fleet inventory management, user account oversight, booking overrides, and revenue reporting.

- **Module 6 – Google Maps/Branch Locator Module**
- Integrates the Google Maps Platform API to show branch locations and uses the Distance Matrix API to calculate drop-off fees between different branches.

- **Module 7 – Notification Module**
- Send automated emails for booking confirmations, cancellation notices, and return reminders.

4. User Interaction Flow

- **The diagrams below trace the step-by-step paths each role follows through the system, from entry point to goal completion**
- **Guest Flow**

Step 1	**■ Land on Home Page**
Step 2	**→ Browse Vehicle Catalog (filter by type / price / fuel)**
Step 3	**→ Open Vehicle Detail Page**
Step 4	**→ Click 'Book Now' → redirected to Login / Register**
Step 5	**→ Complete registration → automatically continues to booking**

Returning Customer Flow

Step 1	■ Log in → redirected to Customer Dashboard
Step 2	Click 'New Booking' → select vehicle & dates
Step 3	→ Review cost summary → confirm reservation
Step 4	→ Receive automated email confirmation
Steo 5	→ Manage, view, or cancel reservation from My Reservations

- **Administrator Flow**

Step 1	**■ Admin login → Admin Dashboard (KPI overview)**
Step 2	**→ Navigate to Fleet Management → add / update vehicles**
Step 3	**→ Navigate to Booking Management → override statuses if needed**
Step 4	**→ Navigate to User Management → manage accounts**
Step 5	**→ Generate revenue report → export data**

3) Database Planning

1. Users Table

- **User_ID** (Primary Key)
- **First_Name**
- **Last_Name**
- **Email** (Unique)
- **Password_Hash**
- **Role** (Guest, Customer, Administrator)
- **Driver_License_Number** (Nullable for Guests/Admins)
- **Account_Created** (Timestamp)

2. Vehicles Table

- **Vehicle_ID** (Primary Key)
- **Make**
- **Model**
- **Year**
- **Category** (e.g., SUV, Luxury, Economy)
- **Fuel_Type**
- **Daily_Rate**
- **Current_Status** (Available, Rented, Maintenance)
- **Current_Branch_ID** (Foreign Key linking to the Branches table)

3. Branches Table

- **Branch_ID** (Primary Key)
- **Branch_Name**
- **Street_Address**
- **City**

- **State**
- **Zip_Code**
- **Latitude** (Crucial for the Maps API integration)
- **Longitude** (Crucial for the Maps API integration)

4. Reservations Table

- **Reservation_ID** (Primary Key)
- **User_ID** (Foreign Key linking to the Users table)
- **Vehicle_ID** (Foreign Key linking to the Vehicles table)
- **Pickup_Date**
- **Dropoff_Date**
- **Pickup_Branch_ID** (Foreign Key linking to the Branches table)
- **Dropoff_Branch_ID** (Foreign Key linking to the Branches table)
- **Total_Cost**
- **Booking_Status** (Active, Completed, Cancelled)

Basic Relationships

- **One-to-Many:** A single User can have multiple Reservations (Booking History).
- **One-to-Many:** A single Vehicle can have multiple Reservations over time.
- **One-to-Many:** A single Branch can have multiple Vehicles currently located there.
- **One-to-Many:** A single Branch can be associated with multiple Reservations as either the pickup or drop-off location.

4)Branding Design

1. System Name Lone Star Rides Rentals Management System (LSR).

2. Logo



3. Color Palette

The color scheme is extracted directly from the primary logo to maintain a cohesive brand identity across the system. The color palette is Deep Navy Blue and Dusty Gold

4. Visual Style Direction

The overall visual style will blend a rugged, off-road aesthetic with a clean, country web layout. To complement the Western styling and the heavy-duty vehicles in the fleet, the primary interface will utilize a dark mode or black theme. This dark background will make the high-quality photos of the vehicles pop, while also providing a comfortable, premium viewing experience for users browsing the catalog

5) wireframes or Mockups

